Chen Sun

csun8109@usc.edu | github.com/ChenSun-YP | linkedin.com/in/chen-sun-79671b134 Site: chensun-yp.github.io

EDUCATION

University of Southern California **Los Angeles, CA**
Ph.D. student, Biomedical Engineering Sept 2023-April 2028
Advisor: Dr. Dong Song

Queens's University **Ontario, Canada**
B.S. degree: Computer Science Sept 2018-May 2022

SKILLS

Python, MATLAB, SQL, Go, PyTorch, OpenCV, ROS, COMSOL; time-series analysis, signal processing, spiking neural networks, nonlinear and nonstationary modeling, reinforcement learning, deep learning, electromagnetic simulation.
Familiarity with Transformers, feedforward neural networks, ResNets, JEPA, and VQ-VAE.
Agentic AI workflows using OpenAI Codex, Claude Code, and Grok.

ACADEMIC EXPERIENCE

Neural Modeling and Interface Laboratory, USC **Los Angeles, CA**
Research Assistant Sept 2023-Present

- Led DARPA INSPIRE project: Identify synaptic plasticity learning rules from endogenous time-series recordings in behaving animals.
    - Built PyTorch model to directly infer population-level LTP/LTD from hippocampal time-series recorded during cognitive learning tasks.
    - Built analysis pipeline for preprocessing and processing raw neural data.
- Led electromagnetic simulation of a research project on how miniaturized TMS interacts with brain tissue and affects cognition in behaving rats.
    - COMSOL finite-element simulation from CT/MRI-derived rat anatomy with tissue-specific electromagnetic properties including field strength, focality, and penetration and tested anatomical variability across different head/brain volumes.
    - Published in Journal of Neural Engineering (2026).

Department of Biomedical Engineering, USC **Los Angeles, CA**
Teaching Assistant Sept 2023-Present

- BME 501-Intro to Biomedical Engineering, BME 513-Signal and System Analysis, BME 552-Neural Implant Engineering

Machine Intelligence & Biocomputing Laboratory (MIB), Queen's University **Ontario, Canada**
Research Assistant March 2022-April 2023

- Investigated the application of various evolutionary algorithms for pruning deep spiking neural networks (DSNNs).
- Researched the use of coevolution algorithms for optimizing DSNNs with multi-objective models.
- Examined the integration of dendrites into SNN models and developed a continuous learning capable dendritic DSNN model.

PUBLICATIONS AND MANUSCRIPTS

Google Scholar

Jiang, W., Choi, W., Hsu, C., Lau, I., McHugh, M., **Sun, C.**, et al. (2026). Chronic subthreshold rTMS transiently disrupts learning but enhances memory and flexibility in healthy rats. *Journal of Neural Engineering, 23(5), 056035.*
Choi, W., Pinto, I. I. L. D., **Sun, C.**, Gupta, G., Song, D., & Bogdan, P. (2026). A Generative Model of Complex Networks Using Graphons and Neural Inverse Operators. arXiv:2610.02439 *(preprint).*
Sun, C., Ren, B., Robinson, B. S., Moore, B., Zhang, X., & Song, D. Identification of Long-Term Synaptic Plasticity from Endogenous Spikes During Learning. *Under submission.*

PROFESSIONAL EXPERIENCE

ByteDance **Beijing, China**

Full-Stack Developer Intern May 2021-Sept 2021

Developed backend systems for two internal platforms:

- Form-Filling Platform: Designed with Python's Django framework and Gunicorn for concurrency, enabling secure data collection from isolated informants via Lark app and online Excel integration, supporting over 100 active users.
- HR Management Platform (HR+): Built a Golang/Hertz admin module for internal operations and an internal transfer platform, deployed company-wide across all HR departments.

Mesoor AI **Shanghai, China**

Web Crawler Developer Intern Aug 2020-Feb 2021

- Designed and developed a modular recruitment recommendation system to automate resume collection and job matching, while implementing anti-crawling strategies to bypass website restrictions.
- Improved BERT resume analysis and targeted market-data crawlers to boost platform accuracy and efficiency.
- Built a post-promotion service and secure proxy generators for scalable data requests and user engagement.

SELECTED PROJECTS

LitAgent: Agentic Literature Intelligence Platform **Los Angeles, CA**

Personal Project Apr 2026-Present

- Built LitAgent , an end-to-end RAG-based literature assistant that ingests Zotero libraries, indexes papers, and delivers grounded QA with citations using user-provided LLMs as agents. Developed interactive wiki and relation-map interfaces with retrieval, graph construction, and orchestration modules for cross-paper synthesis.

GazeReader: Apple Vision Pro PDF Reader App **Los Angeles, CA**

Personal Project Nov 2025-Dec 2025

- Developed GazeReader, immersive ARVR application on visionOS using Swift, RealityKit, and ARKit.
- Implemented spatial PDF viewer with real-time highlighting of read text sections via eye-tracking and hand gestures.

EEG Foundation Challenge **Los Angeles, CA**

Deep Decoding Model for Cross-Task Cross-Subject EEG Decoding Sept 2025-Dec 2025

- Developed PyTorch model to predict behavioral performance metrics from EEG data.
- Incorporated subject- and session-specific embeddings to model individual variability.
- Designed neural masking mechanism: session embedding passed through 2-layer MLP to generate per-session neural mask, aggregating subject-specific channels into a composite neural signal representing overall EEG features.
- Ranked top 10th percentile among all participating teams.

Massachusetts Institute of Technology **Cambridge, MA**

Thalamic Model for Task Representation and Segmentation in Dynamic Environments June 2023-Present

- Built a PyTorch thalamic model for dynamic task segmentation with self-generated internal task representations.
- Trained in a custom OpenAI Gym environment with volatility and task switching to study task-boundary inference.
- Showed single unsegmented network could learn opposing tasks, achieve lifelong learning, and adapt to novel tasks without catastrophic forgetting
- Predicted task-segmentation triggers in animal behavior; compared synaptic plasticity with neural-activity updates.

Queens's University **Ontario, Canada**

EvoMedley, Interactive Melody Generator with Multi-song Fusion Jan 2022-April 2022

- Proposed EvoMedley, an interactive music mixing method using evolutionary algorithms (EA) to generate personalized music from user-selected playlists.
- Developed MIDI feature extraction into minimal melodic fragments for an EA-based music-generation gene pool.
- Designed dynamic user-guided fitness for real-time preference adaptation while retaining original track features.

Tangent **Ontario, Canada**

Customized Online Archive May 2022-Sept 2022

- Led Tangent Collective's online archive: recruitment, project management, backend development, and deployment.
- Built a Django/MySQL backend with NGINX/WSGI on AWS EC2, S3 storage, CloudFront CDN, and RDS for data recoverability; used pm2 for stable automated deployment and daemonization.